

Notes: Bernstein, Frydman, and Hilt (JPE, 2026)

The Value of Ratings: Evidence from Their Introduction in Securities Markets

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1 Big picture

- **Question.** Do credit ratings matter because they *transmit information*, or only because modern regulation and investment mandates make them mechanically important? The paper asks this question in a setting where ratings had *no* regulatory role.
- **Setting.** The paper studies the introduction of John Moody’s first letter-graded bond ratings in April 1909. The setting is the New York Stock Exchange market for railroad bonds, which was already large, active, and information-rich.
- **Core challenge.** In modern data, ratings are entangled with regulations, investment mandates, and bond covenants. That makes it very hard to separate an information channel from a demand-shifting regulatory channel.
- **Main historical advantage.** Moody’s 1909 ratings were a genuinely new financial innovation. They were sold in a printed volume, were not tied to regulation, and were largely based on publicly available information.
- **Main result on prices.** When Moody assigned a rating that was *worse than market-implied expectations*, the bond’s yield rose after publication. The paper estimates increases of about 16–25 basis points on an annualized basis over the subsequent year.
- **Main result on liquidity.** Bonds that received ratings experienced a substantial decline in bid-ask spreads, especially among initially illiquid bonds. The interpretation is that ratings reduced information asymmetries and improved market liquidity.
- **Key mechanism.** Ratings appear to have mattered not because they revealed secret accounting data, but because they *packaged* complex information into a coarse, easy-to-interpret hierarchy of letter grades. That packaging helped investors process risk.
- **Heterogeneity.** The effect of negative rating surprises on yields was much smaller for railroads whose boards were connected to elite underwriters. This suggests that banker reputation and monitoring partly substituted for ratings as certification devices.
- **Why this paper matters.** It is one of the cleanest pieces of evidence that ratings can affect markets through information provision alone. More broadly, it shows that the format in which information is presented can matter for price discovery, liquidity, and the creation of safe assets.

2 Data and measurement (key objects)

2.1 Bond transactions

- **Main source for yields.** Weekly closing prices for all railroad bonds traded on the NYSE, collected from the Monday edition of the *New York Times*.
- **Time window.** Two years of weekly data centered on the introduction of Moody’s ratings in April 1909.
- **Main yield sample.** An unbalanced panel of 562 bonds from 67 railroads, restricted to bonds with at least one traded price before and after the ratings release.

- **Rated sample used for surprises.** About 90% of transactions were for bonds rated by Moody. The main surprise analysis uses 454 bonds for 44 railroads.
- **Outcome variable.** The paper computes a yield to maturity for each bond-week observation and uses it as the main price-based measure of perceived risk.

2.2 Market microstructure and liquidity

- **Main source for liquidity.** Original NYSE quotation sheets, which report bid and ask quotes for listed bonds.
- **Frequency and window.** Weekly observations for the 12 Wednesdays before and after the introduction of ratings, yielding 24 weeks of quote data.
- **Main spread sample.** 346 bonds with at least one observed nonmissing bid-ask spread in both the pre- and postperiods.
- **Liquidity measure.** The bid-ask spread is

$$\text{Spread}_{it} = \frac{2(\text{Ask}_{it} - \text{Bid}_{it})}{\text{Ask}_{it} + \text{Bid}_{it}}.$$

- **Descriptive fact.** Rated bonds were much safer and more liquid on average than unrated bonds. Mean spreads were about 1.1% for rated bonds versus 2.9% for unrated bonds, and mean yields were about 4.3% versus 6.2%.
- **Important overlap.** Despite those mean differences, the distributions overlap substantially. This overlap is crucial for the paper’s liquidity design because some rated and unrated bonds look quite similar ex ante.

2.3 Ratings and director information

- **Ratings data.** The assigned bond ratings and the bond-level information underlying those ratings are collected from *Moody’s Analyses of Railroad Investments* (1909).
- **Railroad characteristics used by Moody.** The paper records factor of safety, income per mile, interest per mile, seniority rank, and bond duration.
- **Board connections.** The authors collect director names from the 1909 *Moody’s Manual* and match them to financiers from top underwriting firms, commercial banks, and trust companies.
- **Why board links matter.** These links proxy for banker reputation and monitoring. The paper uses them to test whether ratings mattered less for firms that were already certified by elite financial connections.

2.4 Unobservables and timing

- **What investors already knew.** Before ratings existed, investors had access to annual manuals with rich accounting information. In that sense, Moody’s ratings largely summarized public information rather than disclosing private issuer data.

- **When ratings were constructed.** The ratings were compiled before publication, using data mostly through the railroads' 1908 fiscal year-ends and some updates through January 1909.
- **When ratings were released.** Moody's volume was released on April 23, 1909. This timing matters because the event-study evidence shows effects around release, not during the earlier construction period.
- **Why timing is useful for identification.** If Moody had merely predicted future yield changes correctly, one would expect movements during the construction window. Instead, the paper finds that relative yields change only after publication.

3 Models and empirical specifications (all equations + interpretation)

3.1 Ratings as public partitioning and surprise construction

The paper's central idea is that market yields before April 1909 reveal investors' *ex ante* beliefs about credit risk, while Moody's ratings reveal how his new classification system partitioned those bonds into discrete risk categories.

To formalize this, the authors first compute each railroad's mean prerating bond yield and place railroads into quartiles of the prerating yield distribution. Within each quartile, the bonds had very similar initial yields. They then compare the railroad's median Moody rating to the median rating within that yield quartile.

A railroad is classified as receiving a **negative surprise** if Moody's rating is worse than the quartile's median rating. Symmetrically, some railroads receive positive surprises.

- **Interpretation.** This construction asks whether Moody reclassified a railroad as safer or riskier than the market had implicitly priced it beforehand.
- **Why quartiles are useful.** They create groups of bonds with very similar initial yields, so subsequent relative changes can be interpreted as the market response to Moody's additional categorization.
- **Why the paper focuses on negative surprises.** Negative surprises are more likely to matter because many of them occur for bonds already viewed as somewhat risky, where new bad news about repayment prospects should be more salient.

3.2 Main yield difference-in-differences regression

The main regression for bond yields is

$$y_{ijt} = \alpha_i + \gamma_t + \delta \text{NegSurprise}_j \times \text{PostRatings}_t + \sum_n \rho_n \text{RatingLevel}_{ni} \times \text{Trend}_t + \varepsilon_{ijt}. \quad (1)$$

- **Outcome.** y_{ijt} is the yield to maturity of bond i issued by railroad j in week t .
- **Fixed effects.** α_i are bond fixed effects and γ_t are week fixed effects.

- **Treatment.** $\text{NegSurprise}_j \times \text{PostRatings}_t$ captures whether negative-surprise bonds moved differently after the ratings were released.
- **Trend controls.** Rating-level indicators interacted with a time trend are included because credit spreads were already drifting downward during the recovery from the Panic of 1907.
- **Key coefficient.** δ is the differential postratings effect of receiving a negative surprise.

Economic interpretation. The regression compares bonds that ultimately received the same Moody rating, but whose initial market yields implied different expectations. If ratings transmitted new information, the negative-surprise bonds should see relative yield increases after publication.

3.3 Event-study and trend versions of the yield design

The paper estimates several versions of equation (1):

- an event-study version with week-by-week treatment effects,
- a *donut* specification that focuses on weeks 26–52 after publication,
- a linear trend version that lets the postratings treatment effect accumulate gradually over time.
- **Why a gradual effect is reasonable.** Moody’s volume diffused through the market over time, and investors likely took time to understand its significance. So the paper does not force an instantaneous jump.
- **Main estimates.** The donut specification implies about a 16 bps annualized increase in yields. The preferred trend specification implies about a 21 bps increase over 12 months. Alternative versions range roughly from 16 to 25 bps.
- **Event-study message.** The preperiod is flat, and the differential yield rises only after April 1909, stabilizing around 20 bps after about six months.

3.4 Bond-level surprises and within-firm variation

To address the concern that railroads with negative surprises may differ from other railroads along omitted dimensions, the paper also constructs *bond-level* surprises and exploits within-firm variation.

- Some bonds of the same railroad received worse-than-expected ratings while other bonds of the same railroad did not.
- The paper reestimates the yield regression with bond-level surprise indicators.
- It then adds firm-specific time trends, so identification comes from differential yield changes *within the same railroad* across its bonds.

Interpretation. If the results survive this tighter comparison, they are much less likely to be driven by railroad-level confounds. They do survive: the within-firm results are very similar in sign and magnitude to the baseline estimates.

3.5 Public determinants of ratings and robustness

A key empirical fact is that Moody’s ratings were highly predictable from public information. The paper shows especially strong correlations between ratings and:

- factor of safety,
- prerating yields,
- income per mile.

The yield regressions are therefore reestimated while interacting the postratings time trend with these prerating observables.

- **Why this matters.** If the negative-surprise effect disappears once these controls are added, then ratings would just be mechanically proxying for public observables.
- **What happens.** The estimated effect of a negative surprise remains nearly unchanged.
- **Additional exercise.** The paper also instruments the surprise term with these publicly observable rating determinants and still finds a significant effect on yields.

Interpretation. Ratings mattered above and beyond simple linear combinations of already available accounting and yield information.

3.6 Ratings, reputation, and heterogeneity across firms

The paper tests whether ratings mattered less for firms whose boards were already tied to elite financiers.

A representative specification interacts the negative-surprise trend effect with an indicator for railroads that had many top underwriters on their board.

- **Prediction.** If elite banker connections already mitigated information asymmetry, then ratings should have less incremental value for those firms.
- **Finding.** That is exactly what the data show. For railroads with at least three top underwriters on the board, the negative-surprise effect on yields is largely offset. For firms with no such connections, the yield response is strong.

Interpretation. Banker reputation and monitoring acted as a substitute for ratings as a certification technology.

3.7 Liquidity regression and the instrumental-variables design

To study the effect of being rated on liquidity, the paper estimates

$$\text{Liquidity}_{it} = \alpha_i + \gamma_t + \nu_1 \text{Rated}_i \times \text{PostRatings}_t + \nu_2 \text{Rated}_i \times \text{Trend}_t + \beta X_{it} + e_{it}. \quad (2)$$

The outcome is usually the bid-ask spread. Because rated bonds were selected rather than random, the paper instruments for the rated-bond terms using the average prerating yields of the *other* bonds issued by the same railroad.

Let $\bar{y}_{-i}$ denote the average prerating yield of the issuer’s other outstanding bonds. Then the key first-stage instruments are

$$\bar{y}_{-i} \times \text{PostRatings}_t \quad \text{and} \quad \bar{y}_{-i} \times \text{Trend}_t.$$

- **Historical logic.** Moody rated all bonds of railroads that had at least some high-quality, low-yield issues of broad investor interest. Therefore, a lower average yield on the *other* bonds of the same railroad made a given bond more likely to be rated, even if that specific bond was relatively illiquid or risky.
- **Main OLS result.** Rated bonds saw spreads fall by about 60 bps relative to unrated bonds after publication.
- **Main 2SLS result.** The IV estimate is much larger—about 310 bps—because the local average treatment effect applies to marginal, initially illiquid bonds for which rating status was most sensitive to the issuer’s other bonds.

Economic interpretation. Ratings mattered most for the bonds with the worst initial information frictions. Those are precisely the bonds for which a simple external signal could do the most to reduce adverse selection.

3.8 Illiquid-bond subsamples and interpretation of the LATE

The paper then restricts the sample to bonds above the 60th and 80th percentiles of prerating illiquidity.

- The estimated spread declines remain very large in absolute value—about 2.9 and 2.6 percentage points, respectively.
- First-stage strength rises sharply in these subsamples.
- The authors interpret this as evidence that the compliers are mainly *marginal, illiquid bonds*: very liquid bonds would have been rated no matter what, but illiquid bonds could go either way depending on whether the issuer also had more prominent securities.

Interpretation. The strongest liquidity benefits of ratings accrued where information asymmetry was initially most severe.

3.9 Trade-size regressions and small-investor participation

Using the same IV design, the paper studies whether ratings changed trading patterns.

- There is no clear effect on the probability that a bond trades on a given day.
- Ratings do increase the number of single-lot trades and the probability of at least one single-lot trade conditional on any trade occurring.
- There is no similar effect on large trades.

Interpretation. Ratings may have encouraged participation by smaller investors in otherwise illiquid securities, consistent with ratings reducing the informational advantage of more sophisticated traders.

4 Identification and instruments (what makes the estimates credible)

4.1 Why modern settings are hard to interpret

- In modern bond markets, ratings affect regulations, investment mandates, and covenants.
- A rating change can therefore move prices even if it contains no information, simply because it changes expected demand from regulated investors.
- The historical setting is valuable precisely because Moody's 1909 ratings had no such regulatory role.

4.2 What identifies the yield effects

- **Unexpected innovation.** The ratings were a new and unanticipated product.
- **Historical timing.** The volume was released on April 23, 1909, after ratings had already been constructed. The market response occurs after release, not during the construction period.
- **Within-yield-quartile comparisons.** Surprises are defined relative to prerating yields, so the treatment compares bonds that looked similar to the market before ratings.
- **Bond and week fixed effects.** These absorb permanent bond differences and common market-wide movements.
- **Rating-level trends.** These absorb the ongoing decline in yields that differed systematically across eventual rating classes.
- **Pretrends.** The event-study evidence shows flat relative trends before release.
- **Within-firm evidence.** Bond-level surprise estimates with firm-specific trends show that the result does not rely on differences across railroads.

4.3 What identifies the liquidity effects

- **Problem.** Rated bonds were safer and more liquid before ratings, so OLS comparisons between rated and unrated bonds are selected.
- **Instrument.** The average prerating yields of the issuer's *other* bonds predict whether a given bond gets rated.
- **Institutional rationale.** Moody rated all bonds of railroads with at least some low-yield, prominent issues, and marginal low-quality issues were carried along into the rated set.
- **Exclusion logic.** After controlling for the focal bond's own prerating yield and spread and allowing for trend differences, the other bonds' average yield should matter for the focal bond's postratings liquidity only through the probability of being rated.
- **Falsification.** The reduced-form instrument does not predict liquidity changes among very liquid bonds, which almost always got rated anyway.

4.4 Additional credibility checks

- The yield effects are robust to controlling for factor of safety, income per mile, interest per mile, duration, prerating yields, and prerating spreads.
- Alternative specifications that compare bonds with similar initial yields but different ratings give similar results.
- There is no evidence that contemporaneous railroad-specific news explains the main yield effects.
- Weak-instrument-robust confidence sets are reported for the IV liquidity regressions.

4.5 Interpretation of what ratings did

- The paper does *not* argue that ratings introduced previously hidden accounting data into the market.
- Instead, the evidence suggests that ratings simplified, standardized, and publicized credit assessment in a form that investors could use more easily.
- The lack of price reversion argues against the interpretation that investors were merely fooled by noisy but salient ratings.
- The hostility of investment bankers to the new ratings supports the idea that ratings eroded informational rents enjoyed by sophisticated intermediaries.

4.6 Relation to the literature

- The paper speaks directly to the modern credit-ratings literature by isolating the pure information channel from regulatory-demand effects.
- It complements work that studies historical ratings quality and determinants by focusing on *market effects*.
- It also connects to work on asymmetric information and liquidity: if ratings reduce the informational advantage of insiders, they should narrow bid-ask spreads.
- More broadly, it contributes to the literature on safe assets and financial innovation by showing how coarse public certification can change market functioning.

5 Tables and figures

5.1 Figure 1: Evolution of US corporate bond markets, 1880–1910

Read:

- The corporate bond market was already large before ratings were introduced: outstanding corporate bonds grew from about \$2 billion in 1880 to about \$15 billion by 1910.
- Railroads dominated the market, but utilities and industrial bonds became more important after 1900.

- NYSE bond trading volume also rose substantially, though trading was concentrated and did not expand as quickly as the stock of bonds outstanding.
- This figure matters because it shows that Moody entered a real, deep securities market rather than inventing ratings for a thin or immature market.

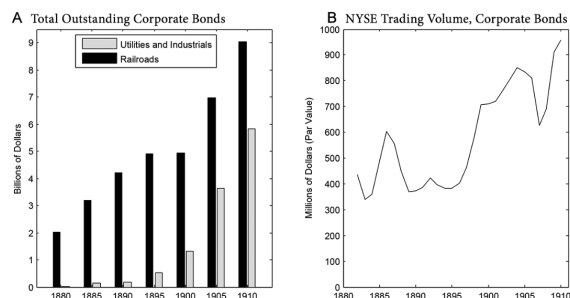


FIG. 1.—Evolution of US corporate bond markets, 1880–1910. *A*, The total value of outstanding corporate bonds, at 5-year intervals, by sector. *B*, A 3-year trailing moving average of total NYSE bond transactions, expressed in millions of dollars of par value. Data for utilities and industrials were compiled from the *Commercial Financial Chronicle's* "Investors' Supplement," for years prior to 1900, and for later years, from Hickman (1952); railroad data are from Hickman (1952). For years after 1900, trading in "railroads and miscellaneous bonds" was recorded from the *New York Times* "Annual Financial Review"; data for earlier years was obtained from Stedman (1905).

TABLE 1
MOODY'S 1909 RATINGS

Rating	Description	Mean			
		Factor of Safety (%)	Income Per Mile (000s)	Seniority Rank (1 = highest)	Yield to Maturity (%)
Aaa	Highest class, value is not affected by any normal changes in earnings capacity of railroad itself	81.25	5.79	6.79	4.29
Aa	While high-grade, slightly inferior to those having first rating in security or in salability	74.11	3.28	8.64	4.24
A	Although high-grade, affected, to partial degree, by changing earning power	67.82	3.16	10.83	4.68
Baa	Generally good but have speculative tinge, good but second-grade issues	50.70	2.67	20.39	4.64
Ba	Make moderately favorable showing and regarded as well secured but affected by changing earning power	56.55	1.34	17.17	5.30
B	More susceptible to fluctuations and regarded as more speculative in position	43.90	1.38	13.22	5.31
Caa	Almost directly responsive to changes in earning power and have not had benefit of available income equal to more	45.00	1.53	27.00	5.18
Ca	Approach more strongly to field of speculative issues with but moderate security	31.04	.97	16.80	7.92
C	Show but a slight margin in surplus above amount required for their interest and which are not well secured	20.00	1.17	16.58	6.88
D	Of doubtful character and almost purely speculative value				
E	Defaulted issues, awaiting the results of reorganizations				

NOTE.—Authors' calculations from data presented in *Moody's Analyses of Railroads Investments*, 1909, and from weekly bond prices reported in the *New York Times*. Average (mean) statistics are computed for each bond-week observation used in the primary regression analysis shown in table 4. "Factor of safety" is the percent of earnings available after interest on the issue (and other issues with equal seniority) has been paid. Income per mile is average earnings available for interest, per mile. Seniority rank is the ranking by Moody of the bond among all issues from the same railroad, with 1 being the most senior bond. The yield to maturity is the average value calculated from closing prices as reported in the *New York Times* over our sample.

5.2 Table 1: Moody's 1909 ratings

Read:

- The table lays out Moody's eleven-grade scale from Aaa to E and describes the intended risk ordering.
- Most railroad bonds were rated A or better, reflecting the relatively safe nature of railroad debt at the time.
- Better-rated bonds had higher factor of safety, higher income per mile, greater seniority, and lower yields.
- So from the very start, the ratings were clearly organized around measurable default risk rather than vague reputational labels.

5.3 Table 2: Pairwise correlations between ratings and firm characteristics

Read:

- Moody's ratings were highly correlated with public fundamentals.
- The strongest simple relationship is with factor of safety, followed by prerating yields and income per mile.
- The basic message is that the ratings were largely reproducible from public data.
- This makes the paper's main result stronger, not weaker: ratings moved markets even though they summarized information that was largely already available.

TABLE 2
RATINGS: PAIRWISE CORRELATIONS WITH FIRM CHARACTERISTICS

Correlations (firm level)	Rating (Aaa = 1)	Factor of Safety	Income per Mile	Interest per Mile	Duration	Bid-Ask Spread	Preperiod YTM (mean)
Rating	1						
Factor of safety	-.82	1					
Income per mile	-.46	.39	1				
Interest per mile	-.08	-.02	.60	1			
Duration	.13	-.11	.03	.44	1		
Bidask spread	.30	-.19	-.17	-.09	.05	1	
Preperiod YTM	.56	-.55	-.43	-.23	-.15	.38	1

NOTE.—Pairwise correlations between firm-level characteristics that may be correlated with Moody's 1909 securities ratings for railroad companies. "Rating" is a simple ordinal ranking of a bond's rating in 1909, with the lowest risk group (Aaa) denoted as 1 and values increasing in increments of 1 from there (e.g., Aa = 2, A = 3, ...,). We calculate the median over all bond-week observations in our sample per firm to obtain a firm-level measure. "Factor of safety" is the percent of earnings available after interest on the issue (and other issues with equal seniority) has been paid. "Average income" is the average earnings available for interest, per mile. "Interest per mile" is the average interest owed per mile of railroad line. "Duration" is the bond's duration given its coupon rate and maturity. "Bidask spread" is the average difference in our sample preratings period between the bid and ask prices from the same day at 11 a.m., when the exchange printed and distributed quotations sheets that included all bid and ask quotations for listed bonds. "Preperiod YTM" is the average yield to maturity calculated from closing prices as reported in the *New York Times* during the months prior to the publication of the ratings volume. Unless otherwise stated, we map bond characteristics to firms by computing the mean value of that variable for the firm across all bond-week observations in our main sample used for our analysis in table 4.

TABLE 3
RAILROAD BOND YIELD QUANTILES

Prerating Yield Quartile	Minimum Yield	Maximum Yield	Mean Yield	Percent Aaa	Percent Aa	Percent A	Percent Baa or Lower	Mean Rating (1 = Aaa)	Median Rating
1	.029	.041	.039	.841	.028	.000	.131	1.421	Aaa
2	.041	.043	.042	.298	.602	.101	.000	1.769	Aa
3	.044	.047	.045	.255	.292	.455	.000	2.188	A
4	.048	.076	.053	.000	.052	.652	.297	3.511	A

NOTE.—Distribution of average yields in the period prior to ratings of bonds of different railroads, sorted into quartiles, and median ratings those railroads' bonds received in each yield quartile. Railroads that received a rating below the median for their quartile are designated as having received a negative surprise. The presented mean yield, distribution of ratings, and mean rating within each yield quartile are computed based on bond-week observations that match the sample used in our prior regression analysis in table 4.

5.4 Table 3: Railroad bond yield quartiles

Read:

- The quartiles sort railroads by their prerating mean yields and then report the distribution of Moody ratings within each quartile.
- The yield ranges inside quartiles are tight, so firms inside a quartile were viewed as fairly similar by the market before ratings.
- Yet there is substantial heterogeneity in ratings within each quartile. For example, even the safest yield quartile contains some railroads that Moody rated noticeably worse than the median.
- That within-quartile rating variation is what makes the surprise design possible.

5.5 Figure 2: Ratings surprises and bond yields

Read:

- The event-study shows essentially no differential pretrend between negative-surprise bonds and other rated bonds before April 1909.
- Immediately after the ratings were released, the relative yield on negative-surprise bonds begins to rise.
- The effect accumulates gradually and stabilizes around 20 basis points after about six months.
- The figure is the paper's clearest visual evidence for a causal effect of ratings on prices.

5.6 Table 4: Main effect of ratings surprises on yields

Read:

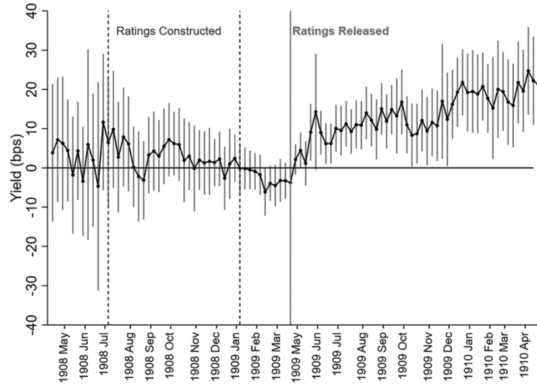


FIG. 2.—Ratings surprises and bond yields. The figure presents the differences in yields between bonds whose ratings constituted a negative surprise and other rated bonds—those that either received a positive surprise or no surprise. The differences plotted are estimated from regressions of yield-to-maturity on indicators for a negative surprise interacted with indicators for each week, with the week prior to the introduction of ratings excluded. The regressions also include bond fixed effects, as well as rating-level fixed effects interacted with time trends (see text). The figure also includes lines representing 95% confidence intervals. The solid vertical line indicates the release of ratings; the dashed lines denote the time period in which Moody constructed the ratings.

Yield (basis points)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Negative surprise $\times$ post	15.50*** (4.87)						
Negative surprise $\times$ trend (weeks) $\times$ post		.47*** (.19)					
Negative surprise $\times$ trend (weeks)		-.04 (.16)					
Negative surprise $\times$ weeks since			.41*** (.12)	.47*** (.12)	.43*** (.13)	.32** (.10)	.35** (.14)
Implied 12-month ATE (bps)	16	24	21	25	22	16	18
95% CI, bps	[6,25]	[15,44]	[9,54]	[12,37]	[7,28]	[3,33]	[4,38]
Bond fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects $\times$ trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm $\times$ trend fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Maturity $\times$ trend fixed effects				Yes			Yes
Level of surprise	Firm	Firm	Firm	Firm	Bond	Bond	Bond
Postperiod duration, weeks	26–52	52	52	52	52	52	52
Postperiod duration, weeks	908	905	905	908	906	917	920
Observations	11,423	15,478	15,478	15,478	15,220	15,220	15,220

NOTE.—Effects of “surprises” (i.e. deviations in ratings from the median for those in the same yield quartile based on their mean yield among all traded bonds prior to the introduction of ratings) on secondary market bond yields trading on the NYSE. Column 1 presents a “dummy” specification; we regress yield to maturity in basis points on a dummy variable equal to 1 for a negative surprise interacted with a dummy variable equal to 1 if the bond transaction occurs 26 weeks after Moody’s securities ratings are released in April of 1909 (post). This specification also includes bond and week fixed effects and rating fixed effects interacted with time trends. The preperiod includes the year prior to the release of ratings, while the post period includes the period 6–12 months following their disclosure. Implied 12-month average treatment effects (ATEs) for the primary coefficient of interest as well as 95% confidence intervals (CIs) for those 12-month estimates are included below the specification. In col. 2, we interact negative surprise with a time trend and then also with a time trend interacted with an indicator variable for the postratings period. Unlike the dummy of col. 1, it includes all 12 months following release of ratings in the post period. In col. 3, negative surprise is interacted with a variable that equals 0 prior to ratings being released and then after the weeks since they were released. Column 4 is the same as col. 3 but also includes 10 bond-level maturity group fixed effects interacted with a time trend. Column 5 is the same as col. 3, but the surprises are calculated at the bond level, rather than the firm level. Column 6 is the same as col. 5 but also includes firm $\times$ trend fixed effects, so that only the within-firm variation is used to estimate the effect of the surprises. Column 7 is the same as col. 6 but also includes 10 bond-level maturity group fixed effects interacted with a time trend. Standard errors clustered at the bond level are in parentheses.

*** Significant at 1%.

** Significant at 5%.

- Across a variety of specifications, negative surprises increase yields by roughly 16–25 basis points over the following year.
- The effect survives maturity controls, bond-level surprise definitions, and specifications using within-firm bond variation.
- The estimates line up closely with the event-study graph, reinforcing the interpretation that the yield response is real and persistent.
- This is the core evidence that ratings transmitted incremental information to the bond market.

5.7 Table 5: Robustness to rating determinants and public observables

Read:

- Adding prerating yields, factor of safety, income per mile, interest per mile, duration, and prerating spreads barely changes the estimated effect of negative surprises.
- Even when surprises are instrumented using public predictors of ratings, the estimated postratings yield response remains significant.
- So the ratings did not matter merely because they loaded on one or two obvious balance-sheet variables.
- A useful way to read this table is: the market responded to the *rating package*, not just to linear combinations of public observables.

5.8 Table 6: Heterogeneity by elite-underwriter board connections

Read:

- The effect of negative surprises on yields is close to zero for railroads with many elite financiers on their board.

	OLS						2SLS
Yield (basis points)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Weeks since ×							
Negative surprise	.41*** (.12)	.41*** (.12)	.54*** (.13)	.54*** (.13)	.54*** (.14)	.53*** (.14)	.64*** (.23)
Prerating yield		4.35 (11.01)				−.651 (13.31)	
Factor of safety			−.006 (.005)	−.006 (.005)	−.006 (.005)	−.007 (.006)	
Average income				−.000004 (.00002)	−.000002 (.00002)	−.000001 (.00002)	
Interest per mile				−.00004 (.0001)	−.00002 (.0001)	−.00002 (.0001)	
Duration					−.004 (.018)	−.006 (.012)	
Prerating bid-ask						5.70 (12.98)	
Implied 12-month ATE (bps)	21	21	18	18	18	17	33

- The effect is large and significant for railroads with no such board connections.
- This suggests that banker ties provided a reputational or monitoring technology that partly substituted for formal ratings.
- The table strengthens the paper's asymmetric-information interpretation: ratings mattered most where private certification was weakest.

Yield (basis points)	All (1)	Many Bankers on Board (2)	Some Bankers on Board (3)	No Bankers on Board (4)
Weeks since ×				
Negative surprise	.54*** (.13)	.011 (.31)	.45*** (.17)	.61*** (.15)
Negative surprise × many top underwriters on board	−.63** (.26)			
All interactions	Yes	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes	Yes
Rating fixed effects × trend (weeks)	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes
Underwriter measure (top 10 bankers on board)	≥3	≥3	[1,2]	0
R ²	.906	.921	.900	.910
Observations	15,478	3,745	7,748	3,985

NOTE.—Heterogeneity in effects of “surprises” (i.e., deviations in ratings from the median for those in the same yield quartile based on their mean yield among all traded bonds prior to the introduction of ratings) on secondary market bond yields trading on the NYSE. Column 1 regresses yield to maturity in basis points on a dummy variable equal to 1 for a negative surprise interacted with a variable that equals 0 prior to ratings being released and then after are the weeks since they were released (weeks since) and interacted with a dummy variable equal to 1 if the firm is in the top quartile of the distribution of connections to elite underwriters, with at least three on their board (many top underwriters on board). This specification also includes bond fixed effects, week fixed effects, and rating fixed effects interacted with weeks since ratings were released. The preperiod includes the year prior to the release of ratings, while the postperiod includes the 12 months following their disclosure. Columns 2–4 estimate the effect of the interaction of negative surprise and weeks since, only among the subset of firms where either more than 2, 1 or 2, or no top underwriters are on the board, respectively. Standard errors clustered at the bond level are in parentheses.

** Significant at 5%.

*** Significant at 1%.

95% CI, hps	[9.34]	[9.34]	[4.32]	[4.31]	[4.32]	[5.31]	[10.56]
Bond fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Rating fixed effects × trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap F-statistic	.905	.905	.905	.905	.905	.905	14.94
Observations	15,478	15,478	15,478	15,478	15,478	15,478	15,478

NOTE.—Effects of “surprises” (i.e., deviations in ratings from the median for those in the same yield quartile based on their mean yield among all traded bonds prior to the introduction of ratings) on secondary market bond yields trading on the NYSE after controlling for other potential observable confounds. Column 1 regresses yield to maturity in basis points on a dummy variable equal to 1 for a negative surprise interacted with a variable that equals 0 prior to ratings being released and then after are the weeks since they were released. This specification also includes bond fixed effects, week fixed effects, and rating fixed effects interacted with weeks since ratings were released. The preperiod includes the year prior to the release of ratings, while the postperiod includes the 12 months following their disclosure. Column 2 is the same as col. 1 but also interacts weeks since with prerating mean yield-to-maturity (prerating yield). Column 3 is the same as col. 1 but also interacts weeks since with a firm's average bonds' factor of safety as a control variable. Column 4 is the same as col. 3 but also interacts weeks since with a firm's bonds' average earnings available to pay interest (average income) and interest per mile as additional control variables. Column 5 is the same as col. 4 but also interacts weeks since with a firm's average bonds' duration as an additional control variable. Column 6 is the same as col. 5 but also interacts weeks since with a firm's bonds' average preratings release bid-ask spread percent and yield-to-maturity as additional control variables. Column 7 uses all control variables in col. 6 but instead as IVs within a 2SLS regression where weeks since × negative surprise is the endogenous variable and yields are again the outcome variable of interest. In this case, CIs are weak instrument robust confidence sets based on the two-step identification robust 95% confidence sets proposed by Andrews (2018) based on linear combination tests, as implemented in the twostepweakiv package in Stata using 1,000 grid points. See also Andrews, Stock, and Sun (2019). All characteristics are computed at the firm level by taking the mean for each bond-week observation in our sample, as in table 2. Standard errors clustered at the bond level are in parentheses.

*** Significant at 1%.

	OLS (1)	2SLS		
		(2)	(3)	(4)
Post × rated issue	−.006* (.003)	−.031*** (.011)	−.029*** (.010)	−.026*** (.010)
Trend × rated issue	.0001 (.0004)	.002 (.001)	.002 (.001)	.002 (.002)
Weak IV CI, post × rated		[−.052, −.010]	[−.049, −.010]	[−.044, −.008]
Bond fixed effects	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes
Week fixed effects × prerating yield	Yes	Yes	Yes	Yes
Week fixed effects × prerating spread	Yes	Yes	Yes	Yes
Preperiod illiquidity (%)			≥60	≥80
Kleibergen-Paap F-statistic		16.4	28.2	60.7
Observations	4,991	4,991	2,036	1,044
Observations not rated (%)	6	6	10	15
Number of bonds	346	346	166	88
Number of bonds not rated	25	25	19	14

NOTE.—Estimates of the effect of ratings on bid-ask spreads, using OLS and the IV approach of eq. (2), estimated via 2SLS. The regressions are estimated using 12 weeks of data before and after the introduction of ratings, for a total of 24 weeks of data. The weak instrument robust confidence sets are two-step identification robust 95% confidence sets proposed by Andrews (2018), based on linear combination tests, as implemented in the twostepweakiv package in Stata using 1,000 grid points. See also Andrews, Stock, and Sun (2019). The prerating yield and the prerating spread are calculated as the mean yield and spread for each issue using all the observations 12 weeks before treatment. Column 1 is an OLS regression of a bond's bid-ask spread on a dummy variable equal to 1 if an issue (bond) is rated interacted with a dummy variable equal to 1 after ratings are released. It also includes bond, week, week × prerating yield, and week × prerating spread fixed effects. Column 2 is the same as col. 1 but uses the IV approach from eq. (2). In particular, after controlling for an issue's yield prior to the release of ratings, we use the yields of other bonds that are part of that same firm in the same period, as an instrument for being rated (because of Moody's propensity to rate all bonds of a given firm). The same-issuer prerating yield is calculated using observations over the 12 weeks before the introduction of ratings. Column 3 is the same as col. 2 but restricts the analysis to only those issues with a mean bid-ask spread at or above the 60th percentile in the period prior to ratings being released. Column 4 is the same as col. 3 but for those above the 80th percentile. The corresponding first stages of the regressions in cols. 2–4 are shown in table A4. Robust standard errors, clustered by bond, are in parentheses.

* Significant at 10%.

*** Significant at 1%.

5.9 Table 7: Effect of being rated on bid-ask spreads

Read:

- OLS suggests rated bonds saw spreads fall by about 60 basis points after ratings were introduced.
- The IV estimates are much larger in magnitude, about 3.1 percentage points in the full sample, because they identify the effect for marginal, illiquid bonds whose rating status was not predetermined.
- When the sample is restricted to bonds above the 60th or 80th percentiles of initial illiquidity, the estimated spread declines remain large and the first stage becomes much stronger.

- The takeaway is that ratings improved liquidity especially for the bonds that suffered the largest information frictions ex ante.

5.10 Table 8: Reduced-form and falsification evidence for the IV strategy

Read:

- The instrument predicts narrower postratings bid-ask spreads in the full sample and, even more strongly, in the illiquid-bond subsamples.
- But it has no meaningful reduced-form effect among the most liquid bonds.
- This is exactly what the identification logic predicts: liquid bonds were nearly always rated anyway, so the instrument should not matter there.
- The table therefore supports both the exclusion restriction and the interpretation that the IV estimates are driven by marginal illiquid securities.

TABLE 8
REDUCED-FORM ESTIMATES

	Reduced Form			Falsification	
	(1)	(2)	(3)	(4)	(5)
Post $\times$ other bonds yields	.353*** (.111)	.394*** (.121)	.385*** (.131)	.080 (.252)	.255 (.212)
Trend $\times$ other bonds yields	-.019 (.016)	-.026 (.020)	-.025 (.023)	-.010 (.014)	-.004 (.008)
Bond fixed effects	Yes	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes	Yes
Week fixed effects $\times$ prerating yield	Yes	Yes	Yes	Yes	Yes
Week fixed effects $\times$ prerating spread	Yes	Yes	Yes	Yes	Yes
Preperiod illiquidity (%)		≥ 60	≥ 80	≤ 20	≤ 60
Kleibergen-Paap F -statistic, first stage	16.4	28.2	60.7	1.0	.5
Observations	4,991	2,036	1,044	1,015	3,010
Observations not rated (%)	6	10	15	1	2
Number of bonds	346	166	88	57	183
Number of bonds not rated	25	19	14	2	6

NOTE.—Estimates of reduced-form relationship between instrument used in estimating eq. (2), yield on other bonds of the same railroads, and bid-ask spreads. The regressions are estimated using 12 weeks of data before and after the introduction of ratings, for a total of 24 weeks of data. The prerating yield and the prerating spread are calculated as the mean yield and spread for each issue using all the observations 12 weeks before treatment. Column 1 is a regression of bid-ask spreads on the instrument, interacted with a postratings indicator and with a time trend. It also includes bond, week, week $\times$ prerating yield, and week $\times$ prerating spread fixed effects. Column 2 is the same as col. 1 but restricts the analysis to only those issues with a mean bid-ask spread at or above the 60th percentile in the period prior to ratings being released. Column 3 is the same as col. 2 but for those above the 80th percentile instead of the 60th. In col. 4, we present the results of a falsification test in which we restrict the sample to the most *liquid* bonds—those in the 20th percentile of prerating spreads or below. In col. 5, we present another falsification test, with the sample restricted to bonds in the 40th percentile of spreads or below. Robust standard errors, clustered by bond, are in parentheses.

*** Significant at 1%.

TABLE 9
IV RESULTS: EFFECT OF RATINGS ON BLOCK SIZE

	Two-Stage Least Squares			
	Any Trade (1)	Number of 1-Lot Trades (2)	1 _{1LotTrade} (3)	Number of Large Trades (≥ 75 th Percentile) (4)
Post April 23, 1909 $\times$ rated issue	.099 (.183)	.574*** (.280)	.428** (.200)	-.104 (.708)
Trend $\times$ rated issue	-.003 (.011)	-.028 (.021)	-.021 (.014)	.031 (.071)
Weak IV robust confidence set,	[.034, 1.113] [.042, .814]			
post $\times$ rated				
Bond fixed effects	Yes	Yes	Yes	Yes
Week fixed effects	Yes	Yes	Yes	Yes
Week fixed effects $\times$ prerating yield	Yes	Yes	Yes	Yes
Week fixed effects $\times$ prerating spread	Yes	Yes	Yes	Yes
Kleibergen-Paap F -statistic	7.9	10.7	10.7	10.7
Dependent variable mean	.277	.403	.333	.663
Observations	8,975	2,418	2,418	2,418
Number of bonds	359	235	235	235
Number of bonds not rated	68	52	52	52

NOTE.—Estimates of effect of ratings on trade size, and in particular on number of lots (of \$1,000 par-value bonds) per transaction. All regressions are versions of eq. (2) estimated via 2SLS, with different second-stage outcomes as shown in the column headings. We use the yields of other bonds that are part of that same firm in the same period as an IV for being rated (because of Moody's propensity to rate all bonds of a given firm). All regressions include bond, week, week $\times$ prerating yield, and week $\times$ prerating spread fixed effects. The prerating yield and the prerating spread are calculated as the mean yield and spread for each issue using all the observations 12 weeks before treatment. Similarly, the instrument, the same-issuer prerating yield, is calculated using observations over the 12 weeks before the introduction of ratings. The regressions are estimated using 12 weeks of data before and after the introduction of ratings, for a total of 24 weeks of data. The weak instrument robust confidence sets are two-step identification robust 95% confidence sets proposed by Andrews (2018) based on linear combination tests, as implemented in the twostepweakiv package in Stata using 1,000 grid points. See also Andrews, Stock, and Sun (2019). Column 1 is a 2SLS regression of the probability of any transaction occurring that day for that bond on an instrumented endogenous dummy variable equal to 1 if an issue (bond) is rated interacted with a dummy variable equal to 1 after ratings are released. Column 2 is the same col. 1, but the dependent variable is the number of standard (single) lot trades in a day, conditional on at least one transaction occurring that day for that bond. Column 3 is the same col. 1, but the dependent variable is a dummy variable equal to 1 if at least one standard (single) lot trade occurs in a day, conditional on at least one transaction occurring that day for that bond. Column 4 is the same col. 1, but the dependent variable is number of large trades in a day, defined as those in the 75th percentile or above, conditional on at least one transaction occurring that day for that bond. The 75th percentile is a 16-lot trade. Robust standard errors, clustered by bond, are in parentheses.

** Significant at 5%.

5.11 Table 9: Effect of ratings on trade size

Read:

- Ratings do not significantly raise the probability that a bond trades at all on a given day.

- But ratings do increase the number and probability of single-lot trades conditional on any trade occurring.
- There is no comparable increase in large trades.
- This pattern is consistent with ratings making illiquid bonds more accessible to smaller investors rather than simply stimulating all trading indiscriminately.

6 Short summary

- **Conclusion.** Moody's first letter-graded ratings changed both prices and liquidity in the bond market even though the ratings had no formal regulatory role.
- **Identification insight.** The paper measures rating surprises relative to prerating market yields and separately uses a railroad-level rating-propensity instrument to identify the liquidity effect of simply being rated.
- **Empirical lesson.** Ratings can matter even when they are largely based on public information, because a coarse and trusted classification system can improve how markets process that information.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that credit ratings can influence security prices and market liquidity through an information channel alone.
- It shows this in perhaps the cleanest possible setting: the first-ever ratings, introduced unexpectedly, before ratings had any regulatory or contractual force.
- The yield evidence says ratings changed investors' beliefs about risk. The liquidity evidence says ratings reduced information asymmetries, especially for illiquid bonds.
- Together, these findings imply that the historical rise of ratings was not merely a legal or regulatory story. Ratings were useful to markets on their own informational merits.

7.2 Economic intuition

- Investors in 1909 already had access to a great deal of information, but that information was costly to process and compare across hundreds of bond issues.
- Moody's innovation converted that complexity into a simple ordinal system that ordinary investors, banks, and institutions could understand quickly.
- Once market participants began to rely on those classifications, bonds with unexpectedly poor ratings required higher yields, while rated bonds—especially marginal, illiquid ones—became easier to trade.

- The weaker response among railroads connected to elite financiers fits the idea that ratings were valuable primarily when other certification mechanisms were missing.

7.3 Takeaway

This paper is best read as a clean historical test of the information role of credit ratings. Its central message is that ratings mattered because they translated complex public information into a common language of risk. That translation changed yields and improved liquidity, showing that information design can matter as much as raw disclosure.